

Surrogate-model comparison — heteroscedastic mixed-variable test problems

Separate GPs / Categorical GP / Standard LVGP (MATLAB) / H-LVGP (MATLAB)

Legend. Metrics on a ground-truth test set (200-pt grid per level for 1-D; 1024 Sobol points per level for Rastrigin 6-D): MAE and RRMSE (surface accuracy), 90% Gneiting–Raftery interval score IS (accuracy + calibration), and 90% coverage (target 0.90; higher is *not* better). **Bold on green** = best model in the comparison group (winners are recomputed within each version’s model set). Budgets n are TOTAL design points; designs are SLHD with n_{rep} replicates per point. Single seed.

Version 1 — all four models

Table 1: Mean-over-levels metrics for every problem $\times$ budget $\times$ model (seed 1). Bold on green = best model in each (problem, budget, n_{rep}) block per metric (MAE/RRMSE/IS: lower; coverage: closest to 0.90).

Problem	n	Model	$n_{\text{rep}} = 10$				$n_{\text{rep}} = 3$			
			MAE $\downarrow$	RRMSE $\downarrow$	IS $\downarrow$	Cov. $\rightarrow .9$	MAE $\downarrow$	RRMSE $\downarrow$	IS $\downarrow$	Cov. $\rightarrow .9$
Branin	10	Separate GPs	26.17	0.8475	360.3	0.6680	26.33	0.8600	349.7	0.6860
		Categorical GP	24.54	0.7582	173.5	0.9420	24.66	0.7623	174.4	0.9420
		Standard LVGP	24.71	0.8273	332.6	0.3300	35.03	1.280	469.1	0.2670
		H-LVGP	43.37	1.663	589.7	0.3540	34.96	1.279	447.5	0.3460
	25	Separate GPs	9.591	0.3852	69.31	0.6860	9.594	0.3704	66.56	0.6710
		Categorical GP	3.582	0.1369	23.57	0.8300	3.611	0.1288	21.75	0.7900
		Standard LVGP	2.553	0.1385	19.18	0.3840	2.769	0.1641	21.31	0.4950
		H-LVGP	2.094	0.0899	11.29	0.9920	1.908	0.1196	13.55	0.9900
	40	Separate GPs	2.980	0.1586	23.32	0.8850	3.201	0.1712	24.54	0.8680
		Categorical GP	0.8582	0.0338	6.284	1.000	1.214	0.0535	5.889	0.9270
		Standard LVGP	1.880	0.0736	19.88	0.1150	1.382	0.0627	19.12	0.2990
		H-LVGP	0.6627	0.0271	9.348	1.000	1.384	0.0555	8.583	0.9800
Six-hump camel	8	Separate GPs	0.9429	1.122	9.221	0.6462	0.9168	1.084	8.743	0.6238
		Categorical GP	0.5557	0.6229	3.890	0.7225	0.5638	0.6158	3.376	0.7613
		Standard LVGP	0.6343	0.7084	8.216	0.2200	0.6275	0.6984	7.442	0.2562
		H-LVGP	0.6319	0.7042	7.014	0.4187	0.6297	0.6964	6.341	0.4050
	20	Separate GPs	0.2411	0.3526	1.524	0.9550	0.2487	0.3837	1.640	0.9575
		Categorical GP	0.1640	0.2511	1.118	0.8525	0.1542	0.2473	0.9751	0.8738
		Standard LVGP	0.2097	0.3159	2.332	0.2725	0.2889	0.4643	4.071	0.1500
		H-LVGP	0.1821	0.2838	1.140	0.9187	0.2537	0.4077	2.191	0.7638
	32	Separate GPs	0.1883	0.3305	1.636	0.8600	0.1984	0.3472	1.804	0.8137
		Categorical GP	0.0845	0.1288	0.5680	0.9287	0.0870	0.1212	0.5249	0.8700
		Standard LVGP	0.1400	0.2662	1.675	0.3425	0.1399	0.2376	1.747	0.2088
		H-LVGP	0.1073	0.2081	0.6652	0.9938	0.0907	0.1342	0.9590	0.9800
Rastrigin 6-D	20	Separate GPs	25.08	1.043	168.8	0.6873	25.19	1.047	169.5	0.6877
		Categorical GP	26.50	1.102	203.5	0.5710	26.59	1.107	204.5	0.5706
		Standard LVGP	30.01	1.233	414.1	0.2427	28.90	1.204	337.2	0.3401
		H-LVGP	29.91	1.229	403.1	0.2583	29.94	1.241	367.7	0.2852
	32	Separate GPs	23.18	0.9554	125.5	0.8306	23.14	0.9538	125.0	0.8264
		Categorical GP	22.90	0.9405	114.7	0.8730	22.77	0.9357	114.6	0.8689
		Standard LVGP	27.35	1.134	255.1	0.5100	27.70	1.150	256.1	0.5103
		H-LVGP	27.27	1.130	249.4	0.5205	27.57	1.145	250.0	0.5208

Table 2: Branin — per-level metrics, $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
10	$x_2 = 15$	25.65	0.4851	323.4	1.000	28.58	0.5221	103.9	1.000	16.74	0.3861	112.3	0.6950	52.82	0.9546	864.8	0.2250
	$x_2 = 2$	28.17	0.9843	305.5	0.8550	25.07	0.8346	210.5	0.8800	25.42	0.7236	238.0	0.4700	29.90	0.9110	200.5	0.6050
	$x_2 = 8$	15.29	1.050	173.9	0.4350	13.48	0.8945	109.9	1.000	18.17	1.243	267.8	0.1350	53.29	3.661	855.1	0.1900
	$x_2 = 0$	47.34	1.116	897.1	0.0800	42.37	0.9837	329.6	0.8450	47.84	1.118	830.0	0.0000	40.37	0.9643	427.8	0.5600
	$x_2 = 10$	14.40	0.6024	101.4	0.9700	13.19	0.5559	113.9	0.9850	15.40	0.6668	214.7	0.3500	40.46	1.824	600.5	0.1900
	<i>Mean</i>	26.17	0.8475	360.3	0.6680	24.54	0.7582	173.5	0.9420	24.71	0.8273	332.6	0.3300	43.37	1.663	589.7	0.3540
25	$x_2 = 15$	5.808	0.1166	37.32	0.5700	3.739	0.0807	19.29	0.8250	3.470	0.0899	38.02	0.5200	3.163	0.0926	16.13	1.000
	$x_2 = 2$	14.27	0.4126	110.1	0.7850	3.752	0.1410	26.61	0.8350	1.699	0.0535	11.19	0.3800	1.618	0.0382	7.714	1.000
	$x_2 = 8$	6.595	0.7796	63.28	0.8750	2.529	0.2223	17.48	0.8600	2.885	0.3188	15.69	0.2350	1.926	0.1685	11.72	1.000
	$x_2 = 0$	16.39	0.2843	106.3	0.2350	5.306	0.1063	32.14	0.8000	1.983	0.0392	14.76	0.5200	1.798	0.0326	8.456	0.9600
	$x_2 = 10$	4.889	0.3331	29.63	0.9650	2.585	0.1341	22.33	0.8300	2.729	0.1911	16.25	0.2650	1.965	0.1176	12.41	1.000
	<i>Mean</i>	9.591	0.3852	69.31	0.6860	3.582	0.1369	23.57	0.8300	2.553	0.1385	19.18	0.3840	2.094	0.0899	11.29	0.9920
40	$x_2 = 15$	0.8645	0.0198	11.03	1.000	1.354	0.0280	7.102	1.000	1.818	0.0483	11.82	0.1400	1.106	0.0334	14.98	1.000
	$x_2 = 2$	5.091	0.2073	46.32	0.8500	0.6727	0.0216	5.507	1.000	2.552	0.0898	31.84	0.0750	0.5152	0.0170	7.361	1.000
	$x_2 = 8$	3.455	0.3626	26.91	0.8650	0.5942	0.0390	6.420	1.000	1.336	0.0973	11.16	0.1100	0.4684	0.0312	8.407	1.000
	$x_2 = 0$	2.275	0.0544	17.26	0.7800	0.4726	0.0092	4.558	1.000	1.788	0.0379	18.36	0.0750	0.3295	0.0074	6.572	1.000
	$x_2 = 10$	3.213	0.1490	15.08	0.9300	1.198	0.0709	7.836	1.000	1.908	0.0946	26.23	0.1750	0.8940	0.0465	9.424	1.000
	<i>Mean</i>	2.980	0.1586	23.32	0.8850	0.8582	0.0338	6.284	1.000	1.880	0.0736	19.88	0.1150	0.6627	0.0271	9.348	1.000

Table 3: Branin — per-level metrics, $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
10	$x_2 = 15$	25.88	0.4893	320.7	1.000	29.56	0.5406	104.7	1.000	38.97	0.7771	530.2	0.3850	38.82	0.7743	520.3	0.4250
	$x_2 = 2$	27.86	0.9856	302.1	0.8600	25.06	0.8329	209.4	0.8850	19.83	0.6723	218.5	0.4850	19.77	0.6726	182.3	0.5700
	$x_2 = 8$	15.69	1.078	169.5	0.4600	13.26	0.8785	113.3	1.000	30.26	1.812	465.1	0.0800	30.18	1.808	447.6	0.2000
	$x_2 = 0$	47.22	1.117	856.5	0.1500	41.85	0.9817	330.5	0.8450	32.82	0.7869	435.2	0.1450	32.81	0.7886	399.3	0.2000
	$x_2 = 10$	15.01	0.6307	99.37	0.9600	13.58	0.5780	113.9	0.9800	53.28	2.353	696.4	0.2400	53.22	2.351	688.1	0.3350
	<i>Mean</i>	26.33	0.8600	349.7	0.6860	24.66	0.7623	174.4	0.9420	35.03	1.280	469.1	0.2670	34.96	1.279	447.5	0.3460
25	$x_2 = 15$	6.429	0.1209	32.81	0.4750	3.802	0.0830	20.12	0.7300	4.002	0.0929	31.27	0.4050	2.343	0.0560	18.15	0.9500
	$x_2 = 2$	14.08	0.4128	110.2	0.7550	3.476	0.1285	23.53	0.8050	1.597	0.0492	9.762	0.4850	1.525	0.0405	10.01	1.000
	$x_2 = 8$	6.514	0.7329	54.17	0.9100	2.974	0.2068	16.39	0.8450	3.221	0.4004	28.72	0.5900	2.444	0.2976	13.77	1.000
	$x_2 = 0$	16.52	0.2858	107.1	0.2250	5.105	0.0989	30.53	0.7000	1.106	0.0195	4.871	0.7850	0.9591	0.0184	11.34	1.000
	$x_2 = 10$	4.426	0.2995	28.59	0.9900	2.701	0.1266	18.15	0.8700	3.920	0.2585	31.95	0.2100	2.269	0.1855	14.47	1.000
	<i>Mean</i>	9.594	0.3704	66.56	0.6710	3.611	0.1288	21.75	0.7900	2.769	0.1641	21.31	0.4950	1.908	0.1196	13.55	0.9900
40	$x_2 = 15$	0.9852	0.0209	10.74	1.000	1.192	0.0215	6.805	0.9000	0.5486	0.0107	7.218	0.4800	2.338	0.0661	15.22	1.000
	$x_2 = 2$	5.359	0.2146	47.05	0.8300	1.213	0.0495	5.608	0.9200	1.939	0.0650	26.36	0.1400	1.210	0.0435	6.854	0.9900
	$x_2 = 8$	4.016	0.3868	28.61	0.8400	0.9340	0.0695	5.523	0.9100	0.9620	0.0730	12.12	0.4150	1.251	0.0835	6.249	0.9500
	$x_2 = 0$	2.266	0.0545	18.01	0.8100	1.088	0.0316	5.163	0.9050	1.251	0.0288	15.29	0.1650	0.7429	0.0183	7.344	0.9600
	$x_2 = 10$	3.380	0.1793	18.31	0.8600	1.644	0.0956	6.347	1.000	2.209	0.1360	34.58	0.2950	1.378	0.0658	7.248	1.000
	<i>Mean</i>	3.201	0.1712	24.54	0.8680	1.214	0.0535	5.889	0.9270	1.382	0.0627	19.12	0.2990	1.384	0.0555	8.583	0.9800

Table 4: Six-hump camel — per-level metrics, $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
8	$x_2 = 0.2$	0.9745	1.360	18.57	0.0700	0.3930	0.5240	2.302	0.8250	0.4574	0.6321	6.801	0.3050	0.4581	0.6349	5.743	0.4350
	$x_2 = 0.4$	0.8588	1.087	3.892	0.8300	0.3520	0.5016	2.132	0.8650	0.6749	0.8368	6.217	0.3300	0.6301	0.7815	4.482	0.5850
	$x_2 = 0.7$	0.7935	0.8747	3.309	0.9900	0.6532	0.7363	4.491	0.6600	0.6573	0.7433	10.22	0.1700	0.6597	0.7446	9.122	0.4000
	$x_2 = 1$	1.145	1.165	11.12	0.6950	0.8244	0.7298	6.636	0.5400	0.7474	0.6216	9.631	0.0750	0.7799	0.6558	8.708	0.2550
	<i>Mean</i>	0.9429	1.122	9.221	0.6462	0.5557	0.6229	3.890	0.7225	0.6343	0.7084	8.216	0.2200	0.6319	0.7042	7.014	0.4187
20	$x_2 = 0.2$	0.1996	0.4172	1.566	0.9250	0.1277	0.2676	0.8634	0.9250	0.3616	0.5417	4.441	0.0750	0.2149	0.3769	1.226	0.8750
	$x_2 = 0.4$	0.3201	0.4179	1.517	0.9700	0.1613	0.2453	0.8614	0.8950	0.1757	0.3015	2.140	0.1900	0.1933	0.2858	1.017	0.8750
	$x_2 = 0.7$	0.1581	0.2428	1.405	0.9550	0.1421	0.2229	1.156	0.8100	0.1463	0.2360	1.541	0.3550	0.1244	0.2066	0.9197	0.9600
	$x_2 = 1$	0.2868	0.3325	1.609	0.9700	0.2250	0.2684	1.589	0.7800	0.1553	0.1843	1.205	0.4700	0.1960	0.2658	1.398	0.9650
	<i>Mean</i>	0.2411	0.3526	1.524	0.9550	0.1640	0.2511	1.118	0.8525	0.2097	0.3159	2.332	0.2725	0.1821	0.2838	1.140	0.9187
32	$x_2 = 0.2$	0.1470	0.4154	1.508	0.9100	0.0670	0.1391	0.4403	0.9700	0.1428	0.4027	1.852	0.3150	0.0944	0.2908	0.6023	1.000
	$x_2 = 0.4$	0.2014	0.3566	1.553	0.9150	0.1006	0.1612	0.6586	0.8800	0.1193	0.2678	1.530	0.2550	0.0949	0.2101	0.5612	0.9900
	$x_2 = 0.7$	0.1841	0.3277	2.077	0.7350	0.0850	0.1225	0.6137	0.8800	0.1258	0.1998	1.216	0.3800	0.1019	0.1819	0.5230	0.9850
	$x_2 = 1$	0.2206	0.2225	1.407	0.8800	0.0856	0.0924	0.5596	0.9850	0.1719	0.1944	2.101	0.4200	0.1380	0.1495	0.9745	1.000
	<i>Mean</i>	0.1883	0.3305	1.636	0.8600	0.0845	0.1288	0.5680	0.9287	0.1400	0.2662	1.675	0.3425	0.1073	0.2081	0.6652	0.9938

Table 5: Six-hump camel — per-level metrics, $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
8	$x_2 = 0.2$	0.9587	1.343	18.44	0.0500	0.4097	0.5318	2.431	0.8800	0.4820	0.6786	6.894	0.3350	0.4803	0.6727	5.995	0.4100
	$x_2 = 0.4$	0.8420	1.062	4.225	0.7350	0.3398	0.4675	1.828	0.9250	0.6352	0.7813	5.313	0.3650	0.6042	0.7421	3.980	0.5550
	$x_2 = 0.7$	0.7615	0.8419	3.532	0.9900	0.6846	0.7519	3.976	0.6750	0.6485	0.7322	9.495	0.1950	0.6589	0.7398	8.700	0.3550
	$x_2 = 1$	1.105	1.089	8.776	0.7200	0.8212	0.7122	5.270	0.5650	0.7443	0.6015	8.065	0.1300	0.7754	0.6310	6.690	0.3000
	<i>Mean</i>	0.9168	1.084	8.743	0.6238	0.5638	0.6158	3.376	0.7613	0.6275	0.6984	7.442	0.2562	0.6297	0.6964	6.341	0.4050
20	$x_2 = 0.2$	0.2209	0.4232	1.602	0.9600	0.2175	0.3302	0.9571	0.7750	0.4742	0.7717	7.023	0.1700	0.4018	0.6536	3.886	0.5750
	$x_2 = 0.4$	0.3670	0.4777	1.698	0.9400	0.1583	0.2668	1.048	0.8350	0.2371	0.4108	3.496	0.0950	0.2519	0.3973	2.006	0.6700
	$x_2 = 0.7$	0.1668	0.2468	1.363	0.9550	0.1234	0.2033	0.9071	0.9250	0.1937	0.3449	2.477	0.1350	0.1593	0.2937	1.332	0.8800
	$x_2 = 1$	0.2401	0.3872	1.896	0.9750	0.1176	0.1889	0.9879	0.9600	0.2507	0.3298	3.290	0.2000	0.2019	0.2862	1.538	0.9300
	<i>Mean</i>	0.2487	0.3837	1.640	0.9575	0.1542	0.2473	0.9751	0.8738	0.2889	0.4643	4.071	0.1500	0.2537	0.4077	2.191	0.7638
32	$x_2 = 0.2$	0.1554	0.4195	1.583	0.9000	0.0772	0.1269	0.3591	0.9300	0.1465	0.3727	1.766	0.0850	0.1004	0.1968	0.8511	1.000
	$x_2 = 0.4$	0.1826	0.3516	1.278	0.7950	0.0897	0.1462	0.5662	0.7950	0.1404	0.2041	1.828	0.1350	0.0645	0.1089	0.7446	0.9350
	$x_2 = 0.7$	0.1814	0.3164	1.794	0.7300	0.0961	0.1260	0.6441	0.7850	0.1021	0.1524	1.100	0.3400	0.0446	0.0814	0.7610	0.9850
	$x_2 = 1$	0.2744	0.3014	2.562	0.8300	0.0851	0.0856	0.5300	0.9700	0.1705	0.2213	2.294	0.2750	0.1532	0.1498	1.479	1.000
	<i>Mean</i>	0.1984	0.3472	1.804	0.8137	0.0870	0.1212	0.5249	0.8700	0.1399	0.2376	1.747	0.2088	0.0907	0.1342	0.9590	0.9800

Table 6: Rastrigin 6-D — per-level metrics, $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
20	level 1	29.04	1.228	163.9	0.7129	25.17	1.070	183.4	0.5957	36.64	1.534	448.4	0.3018	36.62	1.533	446.6	0.3027
	level 2	22.94	0.8990	124.5	0.8193	24.53	0.9679	175.6	0.6279	21.40	0.8397	279.8	0.2754	21.40	0.8400	271.1	0.3018
	level 3	24.95	1.012	250.3	0.4287	28.64	1.163	223.6	0.5215	36.07	1.444	579.4	0.1328	36.03	1.442	568.2	0.1445
	level 4	23.41	1.031	136.5	0.7881	27.65	1.206	231.5	0.5391	25.91	1.116	348.9	0.2607	25.60	1.103	326.3	0.2842
	<i>Mean</i>	25.08	1.043	168.8	0.6873	26.50	1.102	203.5	0.5710	30.01	1.233	414.1	0.2427	29.91	1.229	403.1	0.2583
32	level 1	24.39	1.048	138.3	0.8105	21.56	0.9047	106.8	0.8975	25.82	1.110	241.5	0.5166	25.78	1.109	239.8	0.5205
	level 2	22.64	0.8884	119.9	0.8008	23.23	0.9220	117.5	0.8730	27.73	1.099	257.9	0.4932	27.70	1.098	252.7	0.5049
	level 3	24.75	1.004	139.1	0.7900	22.41	0.9005	112.2	0.8828	27.33	1.126	255.2	0.5312	27.30	1.124	252.5	0.5352
	level 4	20.97	0.8812	104.8	0.9209	24.38	1.035	122.2	0.8389	28.51	1.200	265.6	0.4990	28.28	1.190	252.4	0.5215
	<i>Mean</i>	23.18	0.9554	125.5	0.8306	22.90	0.9405	114.7	0.8730	27.35	1.134	255.1	0.5100	27.27	1.130	249.4	0.5205

Table 7: Rastrigin 6-D — per-level metrics, $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Categorical GP				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
20	level 1	28.90	1.223	159.9	0.7295	25.38	1.080	185.9	0.5957	30.93	1.310	380.7	0.2920	28.92	1.217	350.6	0.2666
	level 2	23.01	0.9022	125.1	0.8203	24.60	0.9711	177.0	0.6270	26.83	1.063	296.3	0.3994	28.62	1.130	355.4	0.3154
	level 3	24.94	1.012	250.4	0.4307	28.45	1.157	220.5	0.5312	27.92	1.138	310.2	0.3467	33.07	1.343	423.0	0.2256
	level 4	23.88	1.051	142.6	0.7705	27.93	1.220	234.7	0.5283	29.92	1.304	361.7	0.3223	29.16	1.276	341.8	0.3330
	<i>Mean</i>	25.19	1.047	169.5	0.6877	26.59	1.107	204.5	0.5706	28.90	1.204	337.2	0.3401	29.94	1.241	367.7	0.2852
32	level 1	24.19	1.040	135.8	0.8145	21.26	0.8928	105.5	0.8936	27.21	1.170	254.0	0.5010	27.13	1.168	252.0	0.5029
	level 2	22.60	0.8870	119.7	0.8018	23.26	0.9238	118.2	0.8682	27.95	1.111	257.3	0.5020	27.89	1.108	250.8	0.5195
	level 3	24.70	1.003	138.8	0.7930	22.19	0.8919	111.4	0.8789	27.62	1.135	259.0	0.5264	27.58	1.134	257.1	0.5283
	level 4	21.08	0.8856	105.5	0.8965	24.38	1.034	123.5	0.8350	28.01	1.185	254.1	0.5117	27.68	1.171	239.9	0.5322
	<i>Mean</i>	23.14	0.9538	125.0	0.8264	22.77	0.9357	114.6	0.8689	27.70	1.150	256.1	0.5103	27.57	1.145	250.0	0.5208

Version 2 — without the Categorical GP

Table 8: Mean-over-levels metrics for every problem $\times$ budget $\times$ model (without the Categorical GP) (seed 1). Bold on green = best model in each (problem, budget, n_{rep}) block per metric (MAE/RRMSE/IS: lower; coverage: closest to 0.90).

Problem	n	Model	$n_{\text{rep}} = 10$				$n_{\text{rep}} = 3$			
			MAE $\downarrow$	RRMSE $\downarrow$	IS $\downarrow$	Cov. $\rightarrow .9$	MAE $\downarrow$	RRMSE $\downarrow$	IS $\downarrow$	Cov. $\rightarrow .9$
Branin	10	Separate GPs	26.17	0.8475	360.3	0.6680	26.33	0.8600	349.7	0.6860
		Standard LVGP	24.71	0.8273	332.6	0.3300	35.03	1.280	469.1	0.2670
		H-LVGP	43.37	1.663	589.7	0.3540	34.96	1.279	447.5	0.3460
	25	Separate GPs	9.591	0.3852	69.31	0.6860	9.594	0.3704	66.56	0.6710
		Standard LVGP	2.553	0.1385	19.18	0.3840	2.769	0.1641	21.31	0.4950
		H-LVGP	2.094	0.0899	11.29	0.9920	1.908	0.1196	13.55	0.9900
	40	Separate GPs	2.980	0.1586	23.32	0.8850	3.201	0.1712	24.54	0.8680
		Standard LVGP	1.880	0.0736	19.88	0.1150	1.382	0.0627	19.12	0.2990
		H-LVGP	0.6627	0.0271	9.348	1.000	1.384	0.0555	8.583	0.9800
Six-hump camel	8	Separate GPs	0.9429	1.122	9.221	0.6462	0.9168	1.084	8.743	0.6238
		Standard LVGP	0.6343	0.7084	8.216	0.2200	0.6275	0.6984	7.442	0.2562
		H-LVGP	0.6319	0.7042	7.014	0.4187	0.6297	0.6964	6.341	0.4050
	20	Separate GPs	0.2411	0.3526	1.524	0.9550	0.2487	0.3837	1.640	0.9575
		Standard LVGP	0.2097	0.3159	2.332	0.2725	0.2889	0.4643	4.071	0.1500
		H-LVGP	0.1821	0.2838	1.140	0.9187	0.2537	0.4077	2.191	0.7638
	32	Separate GPs	0.1883	0.3305	1.636	0.8600	0.1984	0.3472	1.804	0.8137
		Standard LVGP	0.1400	0.2662	1.675	0.3425	0.1399	0.2376	1.747	0.2088
		H-LVGP	0.1073	0.2081	0.6652	0.9938	0.0907	0.1342	0.9590	0.9800
Rastrigin 6-D	20	Separate GPs	25.08	1.043	168.8	0.6873	25.19	1.047	169.5	0.6877
		Standard LVGP	30.01	1.233	414.1	0.2427	28.90	1.204	337.2	0.3401
		H-LVGP	29.91	1.229	403.1	0.2583	29.94	1.241	367.7	0.2852
	32	Separate GPs	23.18	0.9554	125.5	0.8306	23.14	0.9538	125.0	0.8264
		Standard LVGP	27.35	1.134	255.1	0.5100	27.70	1.150	256.1	0.5103
		H-LVGP	27.27	1.130	249.4	0.5205	27.57	1.145	250.0	0.5208

Table 9: Branin — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
10	$x_2 = 15$	25.65	0.4851	323.4	1.000	16.74	0.3861	112.3	0.6950	52.82	0.9546	864.8	0.2250
	$x_2 = 2$	28.17	0.9843	305.5	0.8550	25.42	0.7236	238.0	0.4700	29.90	0.9110	200.5	0.6050
	$x_2 = 8$	15.29	1.050	173.9	0.4350	18.17	1.243	267.8	0.1350	53.29	3.661	855.1	0.1900
	$x_2 = 0$	47.34	1.116	897.1	0.0800	47.84	1.118	830.0	0.0000	40.37	0.9643	427.8	0.5600
	$x_2 = 10$	14.40	0.6024	101.4	0.9700	15.40	0.6668	214.7	0.3500	40.46	1.824	600.5	0.1900
	<i>Mean</i>	26.17	0.8475	360.3	0.6680	24.71	0.8273	332.6	0.3300	43.37	1.663	589.7	0.3540
25	$x_2 = 15$	5.808	0.1166	37.32	0.5700	3.470	0.0899	38.02	0.5200	3.163	0.0926	16.13	1.000
	$x_2 = 2$	14.27	0.4126	110.1	0.7850	1.699	0.0535	11.19	0.3800	1.618	0.0382	7.714	1.000
	$x_2 = 8$	6.595	0.7796	63.28	0.8750	2.885	0.3188	15.69	0.2350	1.926	0.1685	11.72	1.000
	$x_2 = 0$	16.39	0.2843	106.3	0.2350	1.983	0.0392	14.76	0.5200	1.798	0.0326	8.456	0.9600
	$x_2 = 10$	4.889	0.3331	29.63	0.9650	2.729	0.1911	16.25	0.2650	1.965	0.1176	12.41	1.000
	<i>Mean</i>	9.591	0.3852	69.31	0.6860	2.553	0.1385	19.18	0.3840	2.094	0.0899	11.29	0.9920
40	$x_2 = 15$	0.8645	0.0198	11.03	1.000	1.818	0.0483	11.82	0.1400	1.106	0.0334	14.98	1.000
	$x_2 = 2$	5.091	0.2073	46.32	0.8500	2.552	0.0898	31.84	0.0750	0.5152	0.0170	7.361	1.000
	$x_2 = 8$	3.455	0.3626	26.91	0.8650	1.336	0.0973	11.16	0.1100	0.4684	0.0312	8.407	1.000
	$x_2 = 0$	2.275	0.0544	17.26	0.7800	1.788	0.0379	18.36	0.0750	0.3295	0.0074	6.572	1.000
	$x_2 = 10$	3.213	0.1490	15.08	0.9300	1.908	0.0946	26.23	0.1750	0.8940	0.0465	9.424	1.000
	<i>Mean</i>	2.980	0.1586	23.32	0.8850	1.880	0.0736	19.88	0.1150	0.6627	0.0271	9.348	1.000

Table 10: Branin — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
10	$x_2 = 15$	25.88	0.4893	320.7	1.000	38.97	0.7771	530.2	0.3850	38.82	0.7743	520.3	0.4250
	$x_2 = 2$	27.86	0.9856	302.1	0.8600	19.83	0.6723	218.5	0.4850	19.77	0.6726	182.3	0.5700
	$x_2 = 8$	15.69	1.078	169.5	0.4600	30.26	1.812	465.1	0.0800	30.18	1.808	447.6	0.2000
	$x_2 = 0$	47.22	1.117	856.5	0.1500	32.82	0.7869	435.2	0.1450	32.81	0.7886	399.3	0.2000
	$x_2 = 10$	15.01	0.6307	99.37	0.9600	53.28	2.353	696.4	0.2400	53.22	2.351	688.1	0.3350
	<i>Mean</i>	26.33	0.8600	349.7	0.6860	35.03	1.280	469.1	0.2670	34.96	1.279	447.5	0.3460
25	$x_2 = 15$	6.429	0.1209	32.81	0.4750	4.002	0.0929	31.27	0.4050	2.343	0.0560	18.15	0.9500
	$x_2 = 2$	14.08	0.4128	110.2	0.7550	1.597	0.0492	9.762	0.4850	1.525	0.0405	10.01	1.000
	$x_2 = 8$	6.514	0.7329	54.17	0.9100	3.221	0.4004	28.72	0.5900	2.444	0.2976	13.77	1.000
	$x_2 = 0$	16.52	0.2858	107.1	0.2250	1.106	0.0195	4.871	0.7850	0.9591	0.0184	11.34	1.000
	$x_2 = 10$	4.426	0.2995	28.59	0.9900	3.920	0.2585	31.95	0.2100	2.269	0.1855	14.47	1.000
	<i>Mean</i>	9.594	0.3704	66.56	0.6710	2.769	0.1641	21.31	0.4950	1.908	0.1196	13.55	0.9900
40	$x_2 = 15$	0.9852	0.0209	10.74	1.000	0.5486	0.0107	7.218	0.4800	2.338	0.0661	15.22	1.000
	$x_2 = 2$	5.359	0.2146	47.05	0.8300	1.939	0.0650	26.36	0.1400	1.210	0.0435	6.854	0.9900
	$x_2 = 8$	4.016	0.3868	28.61	0.8400	0.9620	0.0730	12.12	0.4150	1.251	0.0835	6.249	0.9500
	$x_2 = 0$	2.266	0.0545	18.01	0.8100	1.251	0.0288	15.29	0.1650	0.7429	0.0183	7.344	0.9600
	$x_2 = 10$	3.380	0.1793	18.31	0.8600	2.209	0.1360	34.58	0.2950	1.378	0.0658	7.248	1.000
	<i>Mean</i>	3.201	0.1712	24.54	0.8680	1.382	0.0627	19.12	0.2990	1.384	0.0555	8.583	0.9800

Table 11: Six-hump camel — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
8	$x_2 = 0.2$	0.9745	1.360	18.57	0.0700	0.4574	0.6321	6.801	0.3050	0.4581	0.6349	5.743	0.4350
	$x_2 = 0.4$	0.8588	1.087	3.892	0.8300	0.6749	0.8368	6.217	0.3300	0.6301	0.7815	4.482	0.5850
	$x_2 = 0.7$	0.7935	0.8747	3.309	0.9900	0.6573	0.7433	10.22	0.1700	0.6597	0.7446	9.122	0.4000
	$x_2 = 1$	1.145	1.165	11.12	0.6950	0.7474	0.6216	9.631	0.0750	0.7799	0.6558	8.708	0.2550
	<i>Mean</i>	0.9429	1.122	9.221	0.6462	0.6343	0.7084	8.216	0.2200	0.6319	0.7042	7.014	0.4187
20	$x_2 = 0.2$	0.1996	0.4172	1.566	0.9250	0.3616	0.5417	4.441	0.0750	0.2149	0.3769	1.226	0.8750
	$x_2 = 0.4$	0.3201	0.4179	1.517	0.9700	0.1757	0.3015	2.140	0.1900	0.1933	0.2858	1.017	0.8750
	$x_2 = 0.7$	0.1581	0.2428	1.405	0.9550	0.1463	0.2360	1.541	0.3550	0.1244	0.2066	0.9197	0.9600
	$x_2 = 1$	0.2868	0.3325	1.609	0.9700	0.1553	0.1843	1.205	0.4700	0.1960	0.2658	1.398	0.9650
	<i>Mean</i>	0.2411	0.3526	1.524	0.9550	0.2097	0.3159	2.332	0.2725	0.1821	0.2838	1.140	0.9187
32	$x_2 = 0.2$	0.1470	0.4154	1.508	0.9100	0.1428	0.4027	1.852	0.3150	0.0944	0.2908	0.6023	1.000
	$x_2 = 0.4$	0.2014	0.3566	1.553	0.9150	0.1193	0.2678	1.530	0.2550	0.0949	0.2101	0.5612	0.9900
	$x_2 = 0.7$	0.1841	0.3277	2.077	0.7350	0.1258	0.1998	1.216	0.3800	0.1019	0.1819	0.5230	0.9850
	$x_2 = 1$	0.2206	0.2225	1.407	0.8800	0.1719	0.1944	2.101	0.4200	0.1380	0.1495	0.9745	1.000
	<i>Mean</i>	0.1883	0.3305	1.636	0.8600	0.1400	0.2662	1.675	0.3425	0.1073	0.2081	0.6652	0.9938

Table 12: Six-hump camel — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
8	$x_2 = 0.2$	0.9587	1.343	18.44	0.0500	0.4820	0.6786	6.894	0.3350	0.4803	0.6727	5.995	0.4100
	$x_2 = 0.4$	0.8420	1.062	4.225	0.7350	0.6352	0.7813	5.313	0.3650	0.6042	0.7421	3.980	0.5550
	$x_2 = 0.7$	0.7615	0.8419	3.532	0.9900	0.6485	0.7322	9.495	0.1950	0.6589	0.7398	8.700	0.3550
	$x_2 = 1$	1.105	1.089	8.776	0.7200	0.7443	0.6015	8.065	0.1300	0.7754	0.6310	6.690	0.3000
	<i>Mean</i>	0.9168	1.084	8.743	0.6238	0.6275	0.6984	7.442	0.2562	0.6297	0.6964	6.341	0.4050
20	$x_2 = 0.2$	0.2209	0.4232	1.602	0.9600	0.4742	0.7717	7.023	0.1700	0.4018	0.6536	3.886	0.5750
	$x_2 = 0.4$	0.3670	0.4777	1.698	0.9400	0.2371	0.4108	3.496	0.0950	0.2519	0.3973	2.006	0.6700
	$x_2 = 0.7$	0.1668	0.2468	1.363	0.9550	0.1937	0.3449	2.477	0.1350	0.1593	0.2937	1.332	0.8800
	$x_2 = 1$	0.2401	0.3872	1.896	0.9750	0.2507	0.3298	3.290	0.2000	0.2019	0.2862	1.538	0.9300
	<i>Mean</i>	0.2487	0.3837	1.640	0.9575	0.2889	0.4643	4.071	0.1500	0.2537	0.4077	2.191	0.7638
32	$x_2 = 0.2$	0.1554	0.4195	1.583	0.9000	0.1465	0.3727	1.766	0.0850	0.1004	0.1968	0.8511	1.000
	$x_2 = 0.4$	0.1826	0.3516	1.278	0.7950	0.1404	0.2041	1.828	0.1350	0.0645	0.1089	0.7446	0.9350
	$x_2 = 0.7$	0.1814	0.3164	1.794	0.7300	0.1021	0.1524	1.100	0.3400	0.0446	0.0814	0.7610	0.9850
	$x_2 = 1$	0.2744	0.3014	2.562	0.8300	0.1705	0.2213	2.294	0.2750	0.1532	0.1498	1.479	1.000
	<i>Mean</i>	0.1984	0.3472	1.804	0.8137	0.1399	0.2376	1.747	0.2088	0.0907	0.1342	0.9590	0.9800

Table 13: Rastrigin 6-D — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 10$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
20	level 1	29.04	1.228	163.9	0.7129	36.64	1.534	448.4	0.3018	36.62	1.533	446.6	0.3027
	level 2	22.94	0.8990	124.5	0.8193	21.40	0.8397	279.8	0.2754	21.40	0.8400	271.1	0.3018
	level 3	24.95	1.012	250.3	0.4287	36.07	1.444	579.4	0.1328	36.03	1.442	568.2	0.1445
	level 4	23.41	1.031	136.5	0.7881	25.91	1.116	348.9	0.2607	25.60	1.103	326.3	0.2842
	<i>Mean</i>	25.08	1.043	168.8	0.6873	30.01	1.233	414.1	0.2427	29.91	1.229	403.1	0.2583
32	level 1	24.39	1.048	138.3	0.8105	25.82	1.110	241.5	0.5166	25.78	1.109	239.8	0.5205
	level 2	22.64	0.8884	119.9	0.8008	27.73	1.099	257.9	0.4932	27.70	1.098	252.7	0.5049
	level 3	24.75	1.004	139.1	0.7900	27.33	1.126	255.2	0.5312	27.30	1.124	252.5	0.5352
	level 4	20.97	0.8812	104.8	0.9209	28.51	1.200	265.6	0.4990	28.28	1.190	252.4	0.5215
	<i>Mean</i>	23.18	0.9554	125.5	0.8306	27.35	1.134	255.1	0.5100	27.27	1.130	249.4	0.5205

Table 14: Rastrigin 6-D — per-level metrics (without the Categorical GP), $n_{\text{rep}} = 3$ (seed 1; MAE/RRMSE/IS lower is better, coverage target 0.90). Bold = best model in the row; the green highlight marks the winners of the *Mean* row.

n	Level	Separate GPs				Standard LVGP				H-LVGP			
		MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9	MAE ↓	RRMSE ↓	IS ↓	Cov. → .9
20	level 1	28.90	1.223	159.9	0.7295	30.93	1.310	380.7	0.2920	28.92	1.217	350.6	0.2666
	level 2	23.01	0.9022	125.1	0.8203	26.83	1.063	296.3	0.3994	28.62	1.130	355.4	0.3154
	level 3	24.94	1.012	250.4	0.4307	27.92	1.138	310.2	0.3467	33.07	1.343	423.0	0.2256
	level 4	23.88	1.051	142.6	0.7705	29.92	1.304	361.7	0.3223	29.16	1.276	341.8	0.3330
	<i>Mean</i>	25.19	1.047	169.5	0.6877	28.90	1.204	337.2	0.3401	29.94	1.241	367.7	0.2852
32	level 1	24.19	1.040	135.8	0.8145	27.21	1.170	254.0	0.5010	27.13	1.168	252.0	0.5029
	level 2	22.60	0.8870	119.7	0.8018	27.95	1.111	257.3	0.5020	27.89	1.108	250.8	0.5195
	level 3	24.70	1.003	138.8	0.7930	27.62	1.135	259.0	0.5264	27.58	1.134	257.1	0.5283
	level 4	21.08	0.8856	105.5	0.8965	28.01	1.185	254.1	0.5117	27.68	1.171	239.9	0.5322
	<i>Mean</i>	23.14	0.9538	125.0	0.8264	27.70	1.150	256.1	0.5103	27.57	1.145	250.0	0.5208